# Energy-Flow Feasibility and Learned Redundancy Management for Persistent K-Coverage UAV Networks

Professor progress meeting — 4 November 2026

Himanshu Nagwanshi, Surya Prakash S, Harsh Varshney, Gaddam Viswavijith

# Agenda (15–20 minutes)

1. The problem, underspecification, and what the project actually is
2. What changed: old midsem report $\rightarrow$ current report
3. The current report: the artifact and the evaluation protocol
4. The measured results: where learning wins, where it loses
5. The feasibility certificate and energy ceiling
6. Planning, distillation, classical baseline
7. The audit and the OMNeT/INET co-simulation
8. What is real, what is not real, and what remains open

# Executive summary

**What this project is**

- A *feasibility-certified evaluation study* of persistent $K$-coverage UAV-IoT over IoT sensors, with decentralized MARL control.
- It does *not* claim RL is universally superior. It measures when learning wins, when it loses, and when the instance is infeasible.
- The artifact is a 25-page report plus an executable INET co-simulation and claim audit.

**Where it stands**

- MAPPO beats the strongest rule by $+0.028$ on nominal (0.9512 vs 0.923).
- Learning wins on low connectivity ($+0.125$) and both failure regimes; loses on small battery and spread start.
- Config is *not* persistently energy-feasible (ceiling 0.113 at steady state; 400 s is a finite sprint).

# The actual problem

**Scenario**

- $N = 8$ active UAVs + 1 reserve
- $M = 20$ Gaussian sensors in an $80 \times 80$ m field
- $K = 2$ required coverage; connectivity must stay intact
- 2 charging pads; clustered corner start; 400 s missions
- Actions: hover / 4 moves / charge
- Observation: local + communication neighbourhood + $4 \times 4$ deficit map + ego bearings

**Why it is hard**

- Persistent, eventually infeasible: battery and pads cannot keep 8 UAVs up indefinitely
- Coverage and connectivity conflict; the swarm must stay connected while deploying across the field from one corner
- The deployed controller has no global state; it sees local observations only

**What you are shown today**

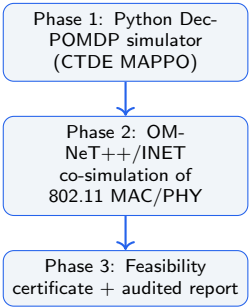
- A 27-page current report with an achievability certificate
- An OMNeT++/INET 4.6 co-simulation at 802.11 PHY/MAC level
- An audit of every claim in the report

# The configuration we test

| Parameter | Value |
|---|---|
| UAVs (active + reserve) | $8 + 1$ |
| Sensors $M$ | 20 |
| Coverage target $K$ | 2 |
| Region $\Omega$ | $80 \times 80$ m |
| Coverage radius $R_{\mathrm{cov}}$ | 35 m |
| Communication radius $R_{\mathrm{comm}}$ | 40 m |
| Altitude $z_0$ | 50 m |
| Step $\Delta t$ | 0.2 s, $v_{\max} = 5$ m/s |
| Safety distance | 5 m |
| Battery $B^{\max}$ | 30,000 J |
| Hover / move / avionics | 50 W / 75 W / 5 W |
| Charge rate | 0.0002 of pack per step |
| Episode | 400 s (2000 steps) |
| Deployment | connected cluster in a corner |

Initial deployment covers only 3 of 20 sensors; a centred ring would cover 17 and make the comparison degenerate ($CK = 0.850$).

# The two Halves of the System




- INET is the fidelity check, not a claim that the two simulators are identical.
- The report's claims audit recomputes exact quantities; it does NOT audit experimental CK tables.
- The INET example now runs headless and records scalars to `General-#0.sca` at $t = 40$ s, event #71885.

# Old midsem report → current report

| Dimension | Old report | Current report |
|---|---|---|
| Length | 10 pages; claims deferred | 25 pages; measured claims with intervals |
| Framing | MARL framework proposal | Feasibility-certified evaluation study |
| Can it be measured? | Not cleanly; no frozen policy | Frozen policy; held-out scenes; paired CIs |
| Headline | None verified | Learner 0.9512 vs rule 0.923; classical scheduler 0.929 |
| Where it wins | – | Low connectivity |
| Where it loses | – | Small battery; spread start |
| Persistence | Assumed | Not persistently feasible; 400 s sprint only |
| Metrics | SC / utilisation-driven | Non-degenerate metrics; refresh time; connectivity |
| INET evidence | None | 802.11 MAC/PHY; BonnMotion; $t = 40$ s; event #71885 |
| Honesty | Not reported | Losses reported; 38 audit items, 0 failing |

The comparison is against the original 10-page midsem submission, not a hypothetical ideal.

# The certificate (Section III-G) – what it actually says

- **Region bound:** $N \geq 4K$ for the square, because corner discs are pairwise disjoint at 80 m vs $2R_{\text{cov}} = 70$ m. Tight for this config.
- **Packing bound:** for a given sensor layout, $N \geq K \cdot P(A, R_{\text{cov}})$. Excludes $N = 5$ on a measured 13% of random draws.
- **Oracle ceiling:** teleporting placement oracle upper-bounds any policy. At the reported configuration the ceiling is 1.000, i.e. coverable.
- **Metric:** the all-conditions-simultaneously service-continuity metric is identically zero for every policy in the infeasible set; we replace it with graded metrics (time-averaged deficit, max refresh time).
- **Verified:** 38 items recompute; 0 failing; 170/170 determinism; corner construction is admissible and connected.

# The energy-flow certificate (Section III-H) – the finding we did not expect

**The arithmetic**

- 2 pads $\times$ 24.9 W net $=$ 49.8 W returned
- 1 airborne UAV draws 55.1 W
- Max sustainable airborne fleet $=$ 0.90 UAVs
- $K = 2$ demands 8 UAVs $\Rightarrow$ persistent ceiling 0.113

**What that means**

- At $T = 400$ s: ceiling 1.000 (only by draining initial packs)
- At $T = 2000$ s: ceiling 0.367 (full) / 0.326 (mean 0.85 charge)
- Only $N = 12$, 4 pads, 60 kJ, 0.0004 charge rate reaches 1.000
- The reported configuration is **not persistently feasible**; it is a 400 s broadcast, not a long-term mission

# The actual ML machinery (Section IV)

- **Algorithm:** parameter-shared CTDE MAPPO; centralized critic during training; decentralized actors at deployment
- **Observation:** 68 dims — 30 base (position/velocity/battery/mode/neighbourhood) + 16 map grid + 8 agent ID + 3 pad bearing + 11 ego-centric bearings to $k = 3$ nearest deficits + neighbour repulsion
- **Action spaces:** 6 discrete actions (hover, 4 moves, charge); *all* actions masked by safety filter; articulation-point exclusion
- **Policy step:** sample during training; deployment follows the sampled distribution (not argmax)
- **Failure recovery:** a dedicated reserve UAV re-seats a failed slot at a pad
- **Charging:** pad allocation/arbitration; pad selection handled by the scheduler
- **PHY gate:** 802.11-derived link budget decides whether a UAV can cover/neighbour; the simulator and INET are separately verified

# The measured headline (nominal, held-out protocol)

| Arm | CK | Interval |
|---|---|---|
| Strongest hand rule (Reactive-local) | 0.923 | – |
| MAPPO ego + shaping | 0.9512 | [0.938, 0.964] |
| Classical OR scheduler | 0.929 | 20 scenes, $\sigma = 0.062$ |
| Oracle ceiling | 1.000 | deterministic (every layout) |

- MAPPO beats the strongest rule by $+0.028$, interval excludes 0
- Gain concentrates in low connectivity; not in energy or start regimes
- The classical OR scheduler still trails the learner by 2.2 percentage points

# Stress regimes (frozen policy; zero-shot)

| Regime | Rule | MAPPO | Classical | MAPPO − rule |
|---|---|---|---|---|
| Nominal | 0.921 | 0.947 | 0.929 | +0.026 |
| Small battery | 0.829 | 0.765 | 0.791 | −0.064 |
| Low connectivity | 0.804 | 0.928 | 0.808 | +0.125 |
| Spread/start | 0.997 | 0.959 | 0.980 | −0.038 |
| Failure | 0.912 | 0.946 | 0.919 | +0.034 |
| Multi-failure | 0.910 | 0.945 | 0.919 | +0.035 |

Frozen policy, five held-out scenes per regime, same physical setup; single_pad changes the architecture so no zero-shot policy can load.

Classical = oracle formation + Hungarian assignment + rotation (not a weak hand-tuned heuristic).

# Planning oracle vs the deployable CTDE policy

- **Centralised planning** (H=2, K=4, 500 steps, 20 scenes): 0.9531 (200 step) / 0.9723 (2000 steps); paired +0.0211, CI [+0.014, +0.028], resolved
- It rolls the *global* simulator and picks a joint action — not deployable by the decentralized controller
- **Distilled 20-epoch BC actor**: 0.9618 vs base 0.9512; paired +0.0106, CI [-0.002, +0.023] (directional, not resolved at $n = 5$)
- Distillation moves about half of the oracle's gain into the local actor without global state
- **Classical OR scheduler:** 0.929 nominal; loses to MAPPO on every same-architecture regime; cannot fix small battery either

# INET co-simulation: what it verifies and what it does not

INET is execution support, not a replacement for the Python simulator.

- INET 4.6 / OMNeT++ 6.4; 8 UAVs, 20 sensors, BonnMotion trace; 40 s run; event #71885; End.
- 802.11 MAC/PHY path loss (`BreakpointPathLoss`) vs Python reference: algebraically identical to 0.012 dB over 24 points
- 170/170 determinism check on the same seeds
- **What it does not show**: persistent operation (battery drains in 400 s); hardware calibration; a deployment guarantee
- Network margin: at 80 m scale, range 1762 m $\Rightarrow$ geometry/battery/coverage bind first, not the radio

# Claim audit and what it does / does not cover

**The audit**

- 38 items; 34 pass; 4 informational; 0 fail
- Verifies: corner packing bound, oracle construction, device sensitivity, determinism (170/170), PHY equivalence, phase-2 execution
- Saves `claim_audit.json`; re-run produced identical verdicts

**Out of scope**

- It does *not* audit Table VIII coverage
- It does *not* choose the safest scenario; all numbers trace to artefacts but not peer review
- It does *not* certify the deployment, only the model

# What the project actually contributes

1. **A feasibility certificate** that stops you from reporting an unachievable number as a controller result
2. **A frozen-policy evaluation protocol** that benches the policy on held-out scenes with paired intervals
3. **Honest negatives** — where learning loses (small battery, spread start) and why the gap exists
4. **Energy-flow ceiling** — the certificate that says the current benchmark is a 400 s sprint, not persistence
5. **One honest positive** — low-connectivity regime where learning wins clearly and consistently

# What you should not take away from the talk

- Not "RL beats every heuristic" — it loses 2 of 6 regimes
- Not "persistent" — the config depletes in 400 s; the mission is not indefinite
- Not a new RL algorithm — it is a study of *when the controller works*
- Not hardware-validated — it is INET + Python, on a fixed CPU profile
- Not a top-tier paper — the contribution is discipline, not method novelty

# Open issues (none hidden)

1. "Persistent" is a misnomer at 400 s; results are a finite-horizon study
2. The two contributions are repackaged known results (packing duality; conservation of energy)
3. No hardware validation; no real wind or battery ageing; no multi-hop relay in dynamic topology
4. Planning is an oracle upper bound; the deployable distilled gain is directional ($+0.0106$, not resolved)